

From Spying To Dynamic Traffic Control With THERMAL CAMERAS



Saurabh Durgopal
is working as
technology
journalist at EFY

The first thermal camera developed for the US army took about an hour to generate a single image. The devices have developed quite a bit since then, and now we have a live feed from thermal imaging cameras. Let us take a look at the additions and developments that these devices have undergone since their inception.

Over the years we have seen many changes in imaging equipment. The time taken to click a thermographic image has come down from an hour to a live video feed of more than 16fps. We also have highly-specialised and portable equipment that can work even in rugged terrains. The optic system and the detector of the camera

that define the sensor have also seen many improvements.

According to T.P. Singh, country manager, FLIR Systems India Pvt Ltd, "Thermal Lepton in a clamp meter or multimeter helps an electrician to look for problems much faster." Lepton is a long-wavelength infrared (LWIR) camera solution that costs less than a dime.

Pixel size has been gradually improving over the years. Now we have 1024×768 pixel modules in the market to cater to high-definition (HD) thermography. Employing such HD modules can significantly improve quality at the cost of making the equipment expensive and increasing power consumption.

"With the introduction of HD video formats even in thermal imaging, a challenge for airports would be setting up of proper communication infrastructure to allow the transfer of all relevant data into a single control centre," says Roy Israely, marketing director, Opgal. As a trade-off, a 640×480 pixel sensor seems like a better option.

Every manufacturer is trying to make the products more user-oriented rather than technology-oriented. This helps developers to go for equipment that provide ease of use. We have manufacturers like FLIR that provide these cameras as an extension to a smartphone. This has made thermal imaging very convenient and popular. Today, we have thermal imaging being used even in television programmes.

According to Israely, "A key factor would be to allow easy integration with various suppliers of visual molecular dynamics and analytics software, allowing designers to implement our products."

Emergence of smartsensors

"A good thermal imager should produce a good thermal image, and this



Fig. 1: Real thermography starts at 160×120 pixels